EXP NO: 11

a) STATIC ROUTING CONFIGURATION USING CISCO PACKET TRACER

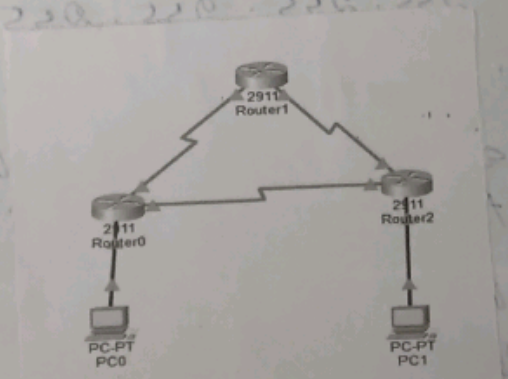
Aim:

To simulate static routing configuration using cisco packet tracer and verify the main and backup routes.

THEORY:

Static routing is a routing method in which routes are manually configured in the routers routing table. A static route specifies the destination network, subnet mask, and next-hop address. Administrative distance (AD) can be used to select a preferred route and configure a backup route. A route with a lower AD is preffered over a route with a higher AD.

Toplogy:



IP Addressing Table:

Device	Interface	IP Address
PC0	Fast Ethernet0	10.0.0.2
Router 0	G0/0	10.0.0.1
Router 0	S0/3/0	20.0.0.1
Router 1	S0/3/0	20.0.0.2
Router 1	S0/3/1	50.0.0.2
Router 2	S0/3/0	50.0.0.2
Router 0	S0/3/1	40.0.0.1
Router 2	S0/3/1	40.0.0.2
Router 2	G0/0	30.0.0.1
PC1	Fast ethernet0	30.0.0.100

Subnet mask:

/8 - 255.0.0.0

/32 - 255.255.255.255

PROCEDURE:

1. Create the network topology using three routers and two PCs in cisco packet tracer.
2. Configure the IP address of the PCs and router interfaces
3. Enable the router interfaces using the no shut down command
4. Configure static routes on Router 0, Router 1 and Router 2.
5. Configure preferred and back up routes using different administrative distance value
6. Verify the routes using show ip route static
7. Test connectivity using Ping
8. Use tracert to verify the path followed by packets

9. Disconnect the main route between routers and Router 1.

10. Verify that the backup route is automatically selected

ROUTING

```
FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.....: FE80::290:2BFF:FE7E:8E4C
   IPv6 Address.....: ::
   IPv4 Address.....: 192.168.1.11
   Subnet Mask.....: 255.255.255.224
   Default Gateway.....: ::
                         192.168.1.1

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.....: ::
   IPv6 Address.....: ::
   IPv4 Address.....: 0.0.0.0
   Subnet Mask.....: 0.0.0.0
   Default Gateway.....: ::
                         0.0.0.0

C:\>ping 192.168.2.11

Pinging 192.168.2.11 with 32 bytes of data:

Reply from 192.168.2.11: bytes=32 time<1ms TTL=127
Reply from 192.168.2.11: bytes=32 time<1ms TTL=127
Reply from 192.168.2.11: bytes=32 time<1ms TTL=127
Reply from 192.168.2.11: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.2.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.3.11

Pinging 192.168.3.11 with 32 bytes of data:

Reply from 192.168.3.11: bytes=32 time<1ms TTL=126
Reply from 192.168.3.11: bytes=32 time=14ms TTL=126
Reply from 192.168.3.11: bytes=32 time<1ms TTL=126
Reply from 192.168.3.11: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.3.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 14ms, Average = 3ms

C:\>ping 192.168.4.11

Pinging 192.168.4.11 with 32 bytes of data:

Reply from 192.168.4.11: bytes=32 time<1ms TTL=126
Reply from 192.168.4.11: bytes=32 time=1ms TTL=126
Reply from 192.168.4.11: bytes=32 time=1ms TTL=126
Reply from 192.168.4.11: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.4.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Result:

Thus static routing was successfully simulated using CISCO PACKET tracer. The main and backup static routes were configured and verified. The backup route was automatically selected when the primary route failed.

b) RIP USING CISCO PACKET TRACER.

Aim:

To implement dynamic routing using RIP in cisco packet router and verify automatic route selection and rerouting.

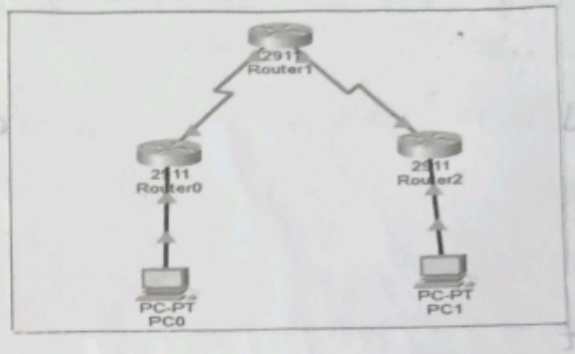
THEORY:

RIP (Routing information protocol) is a distance-vector protocol that uses hop count to select routes. It automatically updates routing information when the network topology changes.

PROCEDURE:

1. Create a network with 3 routers and 2 PCs.
2. Configured IP addresses on PCs and router interfaces.
3. Enabled all required router interfaces using no shutdown.
4. Configured RIP using the router rip command and added the connected networks.
5. Tested connectivity between PC0 and PC1 using Ping
6. Used Tracert to verify the route selected by RIP.
7. Disconnected the direct Router0 – Router2 link.
8. Verified that RIP automatically selected the alternate path through Router1.

TOPOLOGY:




```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.30.1

Pinging 192.168.30.1 with 32 bytes of data:

Reply from 192.168.30.1: bytes=32 time=1ms TTL=255
Reply from 192.168.30.1: bytes=32 time<1ms TTL=255
Reply from 192.168.30.1: bytes=32 time<1ms TTL=255
Reply from 192.168.30.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.30.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.30.3

Pinging 192.168.30.3 with 32 bytes of data:

Reply from 192.168.30.3: bytes=32 time<1ms TTL=128
Reply from 192.168.30.3: bytes=32 time<1ms TTL=128
Reply from 192.168.30.3: bytes=32 time<1ms TTL=128
Reply from 192.168.30.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.30.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.30.4

Pinging 192.168.30.4 with 32 bytes of data:

Reply from 192.168.30.4: bytes=32 time<1ms TTL=128
Reply from 192.168.30.4: bytes=32 time<1ms TTL=128
Reply from 192.168.30.4: bytes=32 time<1ms TTL=128
Reply from 192.168.30.4: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.30.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Result:

Thus, RIP dynamic routing was successfully implemented is cisco packet tracer, and automatic route selection and rerouting were verified